



UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment 3 CLO Mapping: CLO2

Course Code	MAT233	Semester	Spring 2026
Course Title	Linear Algebra		
Instructor	Adnan Rafiq Siddiqui		
Assignment Given Date		Submission Date	
Total Marks	30		

Sr.	Name of student	Registration no.	Section	Assignment marks	Total marks
1					

Q1. Let $V = R^3 = \{(x, y, z); x, y, z \in R\}$ be set of ordered pairs of triples of real numbers with the following operations of addition and scalar multiplication. [10]

$$\vec{u}_1 + \vec{u}_2 = (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2 + 2, y_1 + y_2, z_1 + z_2 - 1)$$

$$k(x_1, y_1, z_1) = (kx_1, y_1, -kz_1).$$

- a. Compute $\vec{u}_1 + \vec{u}_2$ if $\vec{u}_1 = (3, -4, 3)$ and $\vec{u}_2 = (5, -4, 2)$
- b. Compute $k \vec{u}_1$ if $\vec{u}_1 = (-3, -4, 6)$ and $k = -2$
- c. Find zero vector (identity) of the set.
- d. Find additive inverse of the vector $(2, -3, 4)$
- e. Prove or disprove the property $1 * \vec{u}_1 = \vec{u}_1$

Q.2. Determine whether V is a vector space or not. [10]

Let V be the set of all 2×2 matrices with real entries and take the vector space operations on V as the following

$$\vec{u} + \vec{v} = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} + \begin{bmatrix} v_{11} & v_{12} \\ v_{21} & v_{22} \end{bmatrix} = \begin{bmatrix} u_{11} + v_{11} + 1 & u_{12} + v_{12} \\ u_{21} + v_{21} & u_{22} + v_{22} - 3 \end{bmatrix}$$

$$k\vec{u} = k \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} = \begin{bmatrix} k \times u_{11} & k \\ 1 & k \times u_{22} \end{bmatrix}$$

Q.3. Let $\frac{x^2}{7} + \frac{y^2}{15} = 1$ be an ellipse. **Find** the equation of ellipse after shearing it along x-axis by a factor 1.5. Sketch it in GeoGebra and attach its image. **[10]**